

Module 10 Assignment - Arnav Kucheriya

Sunday, March 29, 2026

1. Advantage of Combining Preemptive Priority with Round Robin

Combining Preemptive Priority Scheduling with Round Robin creates a hybrid approach that balances both responsiveness and fairness.

Preemptive Priority ensures that high-priority processes are executed first, improving system responsiveness for critical tasks. However, on its own, it can lead to starvation of lower-priority processes.

By integrating Round Robin within each priority level, processes of the same priority are scheduled in a time-sharing manner, ensuring fairness. This prevents any single process from monopolizing the CPU and guarantees that all processes get execution time.

This combination is especially useful in interactive systems, where:

- High-priority tasks (like user input) need immediate attention
- Lower-priority tasks still require progress

Overall, this hybrid approach improves fairness, responsiveness, and system efficiency compared to using either algorithm alone.

2. Scheduling

Step 1: Gantt Chart

0–2 Idle
2–15 P3
15–21 P1
21–36 P2
36–42 P3
42–48 P1
48–57 P5
57–68 P2
68–80 P5
80–91 P4
91–93 P6

Step 2: Waiting Time

Formula:

Waiting Time = Turnaround Time – Burst Time – I/O Time

Process	Waiting Time
P1	28
P2	0
P3	6
P4	24
P5	30
P6	77

Step 3: Response Time (4 pts)

Formula:

Response Time = First Start Time – Arrival Time

Process	Response Time
P1	7
P2	0
P3	0
P4	24
P5	19
P6	77

Step 4: Turnaround Time (4 pts)

Formula:

Turnaround Time = Completion Time – Arrival Time

Process	Turnaround Time
P1	40
P2	47
P3	40
P4	35
P5	51
P6	79

Notes

- P3 gets priority early (highest priority = 1), but is interrupted by I/O
- P2 dominates mid-execution due to high priority (2)
- P6 suffers the most → lowest priority (6), so it waits longest
- I/O events at t=15 and t=36 correctly push processes into the blocked queue